L2.1 Introduction to Machine Learning Quiz

1. In supervised learning, what information does the teacher provide?
A) Example inputs and their desired outputs
B) Only unlabeled input data
C) Rewards without desired outputs
D) A fixed policy with no examples

ANS:

2. What is the primary goal of unsupervised learning?
A) To predict only continuous values
B) To discover structure or hidden patterns in unlabeled data
C) To maximize cumulative rewards through interaction
D) To compare predictions with teacher-provided labels

ANS:

3. In reinforcement learning, what does an agent attempt to maximize?
A) The number of input features
B) The size of each mini-batch
C) Its cumulative rewards
D) The number of hidden layers

ANS:

4. Which task is an example of regression?
A) Classifying an email as spam or not spam
B) Identifying an image as a cat or dog
C) Predicting whether a statement is true or false
D) Predicting a house price for next year

ANS:

5. Which statement correctly contrasts training and inference?
A) Training adjusts model parameters through repeated forward and backward passes, whereas inference typically uses one forward pass.
B) Training uses one forward pass, whereas inference requires millions of back-propagation iterations.
C) Training always runs on an edge device, whereas inference requires a GPU cluster.
D) Training and inference both update model weights in the same way.

ANS:

6. Which type of value is normally predicted by a classification model?
A) A continuous salary value
B) A discrete class label
C) A model's training time
D) A learning-rate schedule

ANS:

7. What does a loss function measure in supervised learning?
A) The number of layers in a neural network
B) The size of the training dataset
C) How the predicted value differs from the ground-truth value
D) The frequency of an edge processor

ANS:

8. Which function is used in the lecture's logistic-regression model for binary classification?
A) Mean absolute error
B) K-fold averaging
C) Random cropping
D) Sigmoid function

ANS:

9. What layers make up the two-layer fully connected neural network used to solve XOR?
A) One input layer, one hidden layer, and one output layer
B) Two input layers and no output layer
C) One convolution layer and one pooling layer only
D) Five hidden layers and one input layer

ANS:

10. What may happen if a neural network has too many layers and neurons?
A) It must become a regression model.
B) It may overfit the training data.
C) Its model capacity always decreases.
D) It can no longer use activation functions.

ANS:

11. Which loss function is commonly associated with classification?
A) Mean Squared Error only
B) Mean Absolute Error only
C) Cross-Entropy Loss
D) Quantile Loss only

ANS:

12. Which loss function is commonly associated with regression?
A) Focal Loss
B) Hinge Loss
C) Cross-Entropy Loss
D) Mean Squared Error

ANS:

13. In multi-class image classification, what are the two main processing stages?
A) Feature extraction followed by classification
B) Back-propagation followed by data labeling
C) Map construction followed by vehicle control
D) Regression followed by reinforcement learning

ANS:

14. What does the SoftMax output represent in a multi-class classifier?
A) A list of gradients whose values sum to zero
B) A list of class probabilities that sum to 1.0
C) A list containing only ground-truth labels
D) A list of hidden-layer sizes

ANS:

15. In a medical-test example, which outcome is a false positive?
A) A sick person is diagnosed as sick.
B) A healthy person is diagnosed as healthy.
C) A healthy person is misdiagnosed as sick.
D) A sick person is misdiagnosed as healthy.

ANS:

16. In a medical-test example, which outcome is a false negative?
A) A healthy person is correctly diagnosed as healthy.
B) A healthy person is misdiagnosed as sick.

C) A sick person is correctly diagnosed as sick.
D) A sick person is misdiagnosed as healthy.

ANS:

17. What is the purpose of back-propagation during neural-network training?
A) To calculate gradients used to update model parameters
B) To convert a regression problem into clustering
C) To replace the forward pass during inference
D) To divide data into cross-validation folds

ANS:

18. How does mini-batch stochastic gradient descent compute a parameter update?
A) It always uses the entire dataset for every gradient.
B) It uses a small portion of the training data to compute the gradient.
C) It uses validation data as ground-truth test data.
D) It updates parameters without calculating a loss.

ANS:

19. What does a large gap between training accuracy and validation accuracy often imply?
A) The model has no trainable parameters.
B) The model is necessarily underfitting.
C) The model may be overfitting.
D) The validation set has become the training set.

ANS:

20. In 5-fold cross-validation, how is each fold used?
A) All five folds are used only for testing.
B) One fold is permanently discarded.
C) Each fold is used only for training and never for validation.
D) Each fold serves as validation once while the other four serve as training data.

ANS: